

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

January 2019

ALGEBRA (Ph.D. Version)

Instructions to the student

- Answer all six questions; each will be assigned a grade from 0 to 10.
 - Use a different booklet for each question. Write the problem number and your **code number (not your name)** on the outside of the booklet.
 - Keep scratch work on separate pages in the same booklet.
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- (a) Let p be prime and let G be a group of order p^n for some $n \geq 1$. Show that G has a non-trivial center (the center is the set of elements of G that commute with all elements of G).
(b) Suppose G has order p^3 . Show that each conjugacy class in G has order 1 or p . (*Hint*: If two elements are conjugate in G , then they are conjugate in any quotient group of G .)
- Let $n \geq 2$ and let A be an $n \times n$ matrix with complex entries. Prove that the following two statements are equivalent:
(i) A is diagonalizable.
(ii) For every two-dimensional subspace $W \subseteq \mathbb{C}^n$ such that $AW \subseteq W$, the linear transformation given by A restricted to W is diagonalizable.
- Let R be a commutative ring with 1. An R -module F is called *flat* if whenever $f : M \rightarrow N$ is an injective homomorphism of R -modules, the induced map $M \otimes_R F \rightarrow N \otimes_R F$ is also injective. Suppose R is a principal ideal domain. Show that a finitely generated R -module is flat if and only if it is torsion-free.
- Let R be a commutative ring with 1. The ring R is said to be Artinian if every descending chain of ideals stops. That is, for each sequence

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$$

of ideals of R , there exists n such that $I_n = I_{n+1} = \cdots$.

- Show that if R is Artinian and I is an ideal of R , then R/I is an Artinian ring.
- Show that if R is an Artinian integral domain, then R is a field. (*Hint*: Consider the ideals $(x) \supseteq (x^2) \supseteq (x^3) \cdots$.)
- Show that if R is Artinian and P is a prime ideal of R , then P is a maximal ideal.
- Show that if $P_1, \dots, P_n$ are distinct prime ideals of an Artinian ring R , then there exists an element $x \in P_1 \cap \cdots \cap P_{n-1}$ with $x \notin P_n$.
- Show that an Artinian ring has only finitely many prime ideals.

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5. Let $\mathbb{C}(T)$ be the field of rational functions in T over the complex numbers. Let n be a positive integer and let ζ be a primitive n th root of unity.
- (a) Let σ be the automorphism of $\mathbb{C}(T)$ that maps each rational function $f(T)$ to $f(\zeta T)$. Show that the fixed field of σ is exactly $\mathbb{C}(T^n)$. (*Note:* You do not need to show that σ is an automorphism.)
- (b) Find all intermediate fields K between $\mathbb{C}(T^n)$ and $\mathbb{C}(T)$. (Express them in the form $\mathbb{C}(g(T))$ for appropriate rational functions $g(T)$.)
6. Let G be a finite abelian group of order n , let $\hat{G}$ be its set of irreducible complex characters, and let $\mathbb{C}[G]$ be its group ring. Consider a set of indeterminates x_g indexed by the elements of G , and define the matrix $M = (x_{gh^{-1}})_{g,h \in G}$ (the elements of G are given some fixed ordering $g_1, g_2, \dots$).
- (a) Let $\chi \in \hat{G}$. Show that the column vector $v_\chi = (\chi(g_1), \chi(g_2), \dots)^t$ is an eigenvector of M and determine its eigenvalue.
- (b) Show that the vectors v_χ for $\chi \in \hat{G}$ are linearly independent.
- (c) Show that $\det(M) = \prod_{\chi \in \hat{G}} \sum_{g \in G} \chi(g)x_g$.